GNN-Based Genome Scaffolding with Distant and Multiple Reference Assistance

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Introduction

- Current scaffolding methods rely on external long-range data, including long-read sequencing, Hi-C interaction, or closely related reference genomes.
- Here, we propose a novel graph-based deep learning approach that solely leverages intrinsic k-mer distribution data.
- In validation, our model showed good performance compared to the Node2Vec baseline.
- Experiments on challenging datasets covering diverse Mammalian lineages demonstrated its robustness, particularly with distantly diverged species.

Method_ Graph Dataset Construction

- The scaffolder first extracts synteny information from the target genome, with syntenies defined as conserved genomic regions shared across multiple species.
- In the resulting graph, nodes represent the termini (heads or tails) of syntenic blocks, while edges denote the connectivity and directional relationships between these blocks.
- The Hellinger distance is calculated between k-mer distributions for each pair of nodes (syntenies), quantifying their dissimilarity.

Method_ Link Prediction

1. Subgraphs are sampled from the input dataset and fed into the layers of the graph neural network (GNN) model.
2. The model learns patterns indicative of connectivity between node pairs by utilizing node and edge features.
3. The prediction layer outputs the probability of a link existing between nodes.
4. Finally, the resulting graph annotated with updated link probabilities is traversed using a breadth-first search, Dijkstra-like algorithm to reconstruct chromosome-level scaffolds.

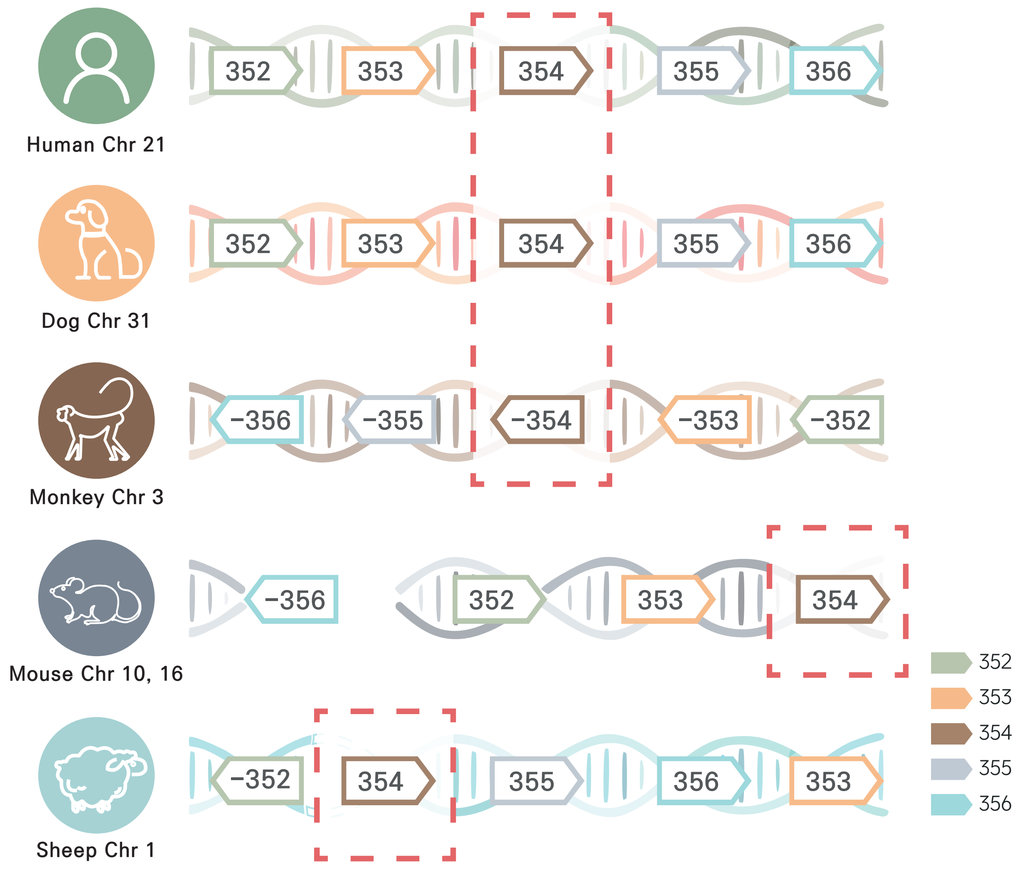


Figure1. Visualization of conserved syntenic regions across 5 Mammalian genomes.

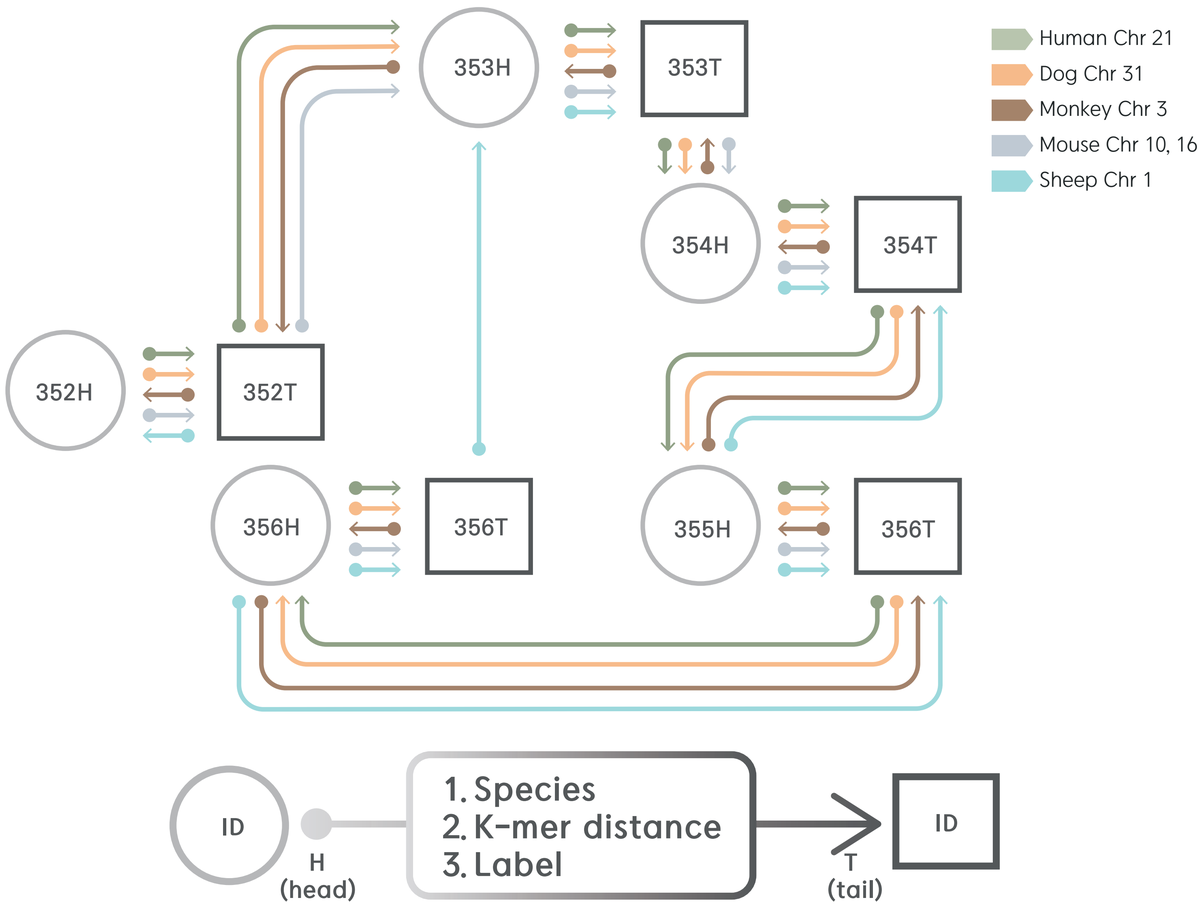


Figure II. Graphical representation of synteny connectivity with node and edge features.

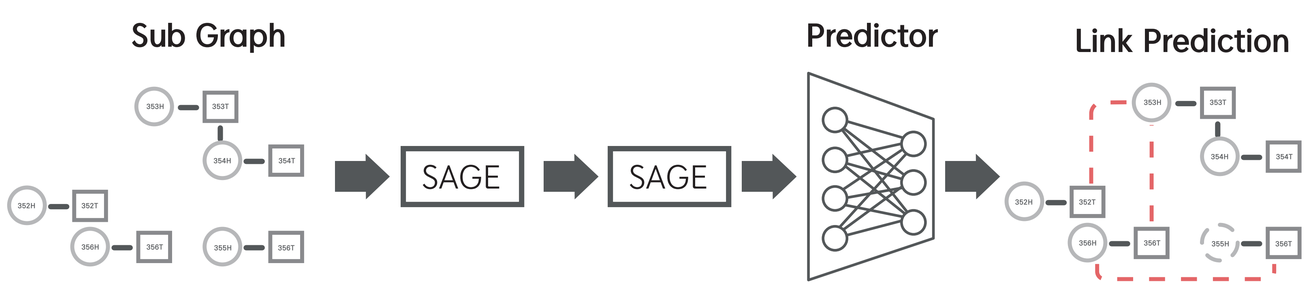


Figure III. Overview of the link prediction process.

Results

- The model was tested with two graph datasets: one with closely related species, and another with distantly related species.
- The primate dataset included 290 syntenies derived from the genomes of bonobo, chimpanzee, gibbon, human, orangutan, and gorilla.
- The Mammalian dataset comprising 674 syntenies derived from dog, mouse, human, monkey, and sheep genomes.
- As a result, GraphSAGE layers achieved the highest F1-scores for both primate and Mammalian datasets (0.8432 and 0.8371, respectively).
- The model showed effectiveness in predicting synteny order.

Table I. Results of performance tests on closely related species

Layer	Seq len	Kmer size	Embed size	Pred. embed.	BLNCD_acc	F1
vanilla	50k	3	16	2	0.515	0.703
SAGE	100k	5	32	8	0.720	0.843
GAT	100k	3	16	8	0.685	0.746

Table II. Results of performance tests on distantly related species

Layer	Seq len	Kmer size	Embed size	Pred. embed.	BLNCD_acc	F1
vanilla	50k	7	32	2	0.499	0.770
SAGE	100k	3	8	2	0.641	0.837
GAT	100k	7	8	2	0.720	0.764

Comparison with Baseline

- As a baseline, we employed Node2Vec, an embedding-based graph machine learning model using random walks.
- Node2Vec performed well on the primate dataset (F1-score: 0.892) but poorly on the Mammalian dataset (F1-score: 0.657).
- This indicates that by effectively incorporating local neighborhood information, the GNN significantly enhances scaffolding link prediction.

Reference

[1]Vrček, Lovro, et al. “Geometric deep learning framework for de novo genome assembly.” Genome research 35.4 (2025): 839-849.